Exp. No: 13

Implement Your Own Ping Program

Aim:

To implement a simple Ping program in computer networks to check whether a destination host is reachable and to measure the round-trip time (RTT) of packets.

Algorithm:

1. Start the program.
2. Read the destination host name or IP address from the user.
3. Create a raw ICMP socket.
4. Construct an ICMP Echo Request Packet containing: ICMP type, Code, Checksum, Identifier, Sequence number, Data.
5. Calculate the checksum of the packet.
6. Send the ICMP Echo ~~Reply~~ Request to the destination.
7. Wait for an ICMP Echo Reply.
8. Calculate the round-trip time between sending the request and receiving the reply.
9. Display the destination, packet size, sequence number and RTT.
10. Repeat the process for a specified number of packets.
11. Display the final result and terminate.

Program:

```
import socket
import struct
import time
import os
def checksum(data):
    total = 0
    for i in range(0, len(data)-1, 2):
        total += (data[i] << 8) + data[i+1]
    if len(data) % 2:
        total += data[-1] << 8
    total = (total >> 16) + (total & 0xffff)
```

```
        total += total >> 16
        return (~total) & 0xffff
def ping(host):
    try:
        dest_addr = socket.gethostbyname(host)
    except socket.gaierror:
        print("Unable to resolve host")
        return
    icmp = socket.getprotobyname("icmp")
    try:
        sock = socket.socket(socket.AF_INET, socket.SOCK_RAW,
                             icmp)
    except PermissionError:
        print("Permission denied. Run as administrator/root.")
        return
    sock.settimeout(2)
    packet_id = os.getpid() & 0xffff
    sequence = 1
    data = b"Computer Networks Ping"
    header = struct.pack("!BBHHH", 8, 0, 0, packet_id,
                         sequence)
    packet = header + data
    checksum = checksum(packet)
    header = struct.pack("!BBHHH", 8, 0, chksum, packet_id,
                         sequence)
    packet = header + data
    start_time = time.time()
    try:
        sock.sendto(packet, (dest_addr, 1))
        sock.recvfrom(1024)
        end_time = time.time()
        rtt = (end_time - start_time) * 1000
```

```
            print ("Reply from ", dest_addr, ": bytes =", len(packet),
                                    "time = %.2f ms" %. rtt)
        except socket.timeout:
            print ("Request timed out")

        finally:
            sock.close()
host = input("Enter hostname or IP address:")
for i in range (4):
        ping (host)
        time.sleep (1)
```

RESULT:

Thus, a Ping Program was implemented successfully using Python and raw ICMP sockets to check destination reachability and measure the round-trip time (RTT) of packets.